Data Set **Descriptions**

In this project you'll query 2 different datasets, intel.device_data and intel.impact_data, that you will join together for your analysis. Here you'll find the data dictionary for each dataset.

intel.device_data

- device_id: Unique identifier for each repurposed device
- device_type: Type of device, values are either "Laptop" or "Desktop"
- model_year: The year the device was manufactured (e.g., 2018, 2019, etc.)

intel.impact_data

- impact_id: Unique identifier for the repurposed device's impact record (e.g., "LP20NA141592")
- device_id: Unique identifier linking the impact record to a specific device in the intel.device_data table
- usage_purpose: The specific purpose for which the device is being repurposed, values are Education & Digital Literacy, Corporate & Enterprise, Government & Public Sector, Environmental Sustainability Programs, and Social Impact & Non-Profit
- power_consumption: Power consumption of the device in watts (W) when in use (e.g., 50W, 75W)
- energy_savings_yr: Estimated energy savings per device per year when repurposed compared to a new device, measured in kilowatt-hours (kWh)
- co2_saved_kg_yr: Estimated CO2 emissions saved per device per year from manufacturing a new device, measured in kilograms (kg).
- recycling_rate: The percentage of the device that is recyclable (e.g., 80%, 90%).
- region: The geographical region where the device was repurposed, values are "North America", "Europe", and "Asia"

```
-- Join device_data and impact_data
SELECT *
FROM intel.device_data d
JOIN intel.impact_data i ON d.device_id = i.device_id;

-- Add device_age column
SELECT d.*, i.*, (2024 - d.model_year) AS device_age
FROM intel.device_data d
JOIN intel.impact_data i ON d.device_id = i.device_id;
```

device_id	device_type	model_year	impact_id	usage_purpose	power_consumption	energy_savings_yr	co2_saved_kg_yr	recycling_rate	region	device_age
LP22AS72961	Laptop	2022	LP22AS7296100004	Corporate & Enterprise	47	25.8	15.5	0.87	Asia	2
LP20EU11904	Laptop	2020	LP20EU1190400006	Education & Digital Literacy	51	24.2	6.1	0.87	Europe	4
LP21NA26999	Laptop	2021	LP21NA2699900006	Corporate & Enterprise	47	5.6	2.3	0.87	North America	3
DT20AS66801	Desktop	2020	DT20AS6680100007	Education & Digital Literacy	78	6.6	4.0	0.98	Asia	4
LP23NA40282	Laptop	2023	LP23NA4028200010	Corporate & Enterprise	43	26.4	10.6	0.87	North America	1
DT22NA80684	Desktop	2022	DT22NA8068400012	Corporate & Enterprise	79	24.7	9.9	0.94	North America	2
LP21EU87824	Laptop	2021	LP21EU8782400013	Corporate & Enterprise	52	37.9	9.5	0.88	Europe	3
LP19AS99290	Laptop	2019	LP19AS9929000022	Education & Digital Literacy	54	58.6	35.2	0.89	Asia	5
LP22NA56098	Laptop	2022	LP22NA5609800023	Corporate & Enterprise	45	23.9	9.6	0.87	North America	2
LP19NA31352	Laptop	2019	LP19NA3135200025	Education & Digital Literacy	55	28.9	11.6	0.89	North America	5
LP20NA34864	Laptop	2020	LP20NA3486400026	Education & Digital Literacy	53	23.0	9.2	0.88	North America	4
DT20NA55932	Desktop	2020	DT20NA5593200026	Corporate & Enterprise	82	38.3	15.3	0.96	North America	4
LP21AS53250	Laptop	2021	LP21AS5325000027	Education & Digital Literacy	50	32.6	19.6	0.89	Asia	3
LP23EU99312	Laptop	2023	LP23EU9931200033	Corporate & Enterprise	45	18.0	4.5	0.89	Europe	1
LP22NA92251	Laptop	2022	LP22NA9225100034	Corporate & Enterprise	48	47.1	18.8	0.87	North America	2
LP21NA30071	Laptop	2021	LP21NA3007100034	Corporate & Enterprise	52	34.9	14.0	0.88	North America	3
LP19AS25034	Laptop	2019	LP19AS2503400036	Corporate & Enterprise	53	23.3	14.0	0.88	Asia	5
LP22EU39620	Laptop	2022	LP22EU3962000037	Corporate & Enterprise	46	13.8	3.5	0.87	Europe	2
LP22AS53174	Laptop	2022	LP22AS5317400037	Education & Digital Literacy	46	-4.9	-2.9	0.88	Asia	2
LP22AS21742	Laptop	2022	LP22AS2174200043	Education & Digital Literacy	46	34.0	20.4	0.86	Asia	2
DT21NA35986	Desktop	2021	DT21NA3598600046	Government & Public Sector	80	31.0	12.4	0.94	North America	3
DT21AS63017	Desktop	2021	DT21AS6301700047	Government & Public Sector	81	28.4	17.0	0.94	Asia	3
DT20NA64068	Desktop	2020	DT20NA6406800048	Corporate & Enterprise	80	19.2	7.7	0.96	North America	4

```
-- Add device_age_bucket column
SELECT d.*, i.*,
    (2024 - d.model_year) AS device_age,
    CASE
        WHEN 2024 - d.model_year <= 3 THEN 'newer'
        WHEN 2024 - d.model_year > 3 AND 2024 - d.model_year <= 6 THEN 'mid-age'
        WHEN 2024 - d.model_year > 6 THEN 'older'
    END AS device_age_bucket
FROM intel.device_data d
```

JOIN intel.impact_data i ON d.device_id = i.device_id;

device_id	device_type	model_year	impact_id	usage_purpose	power_consumption	energy_savings_yr	co2_saved_kg_yr	recycling_rate	region	device_age	device_age_bucket
LP22NA30289	Laptop	2022	LP22NA3028902256	Education & Digital Literacy	48	4.9	1.9	0.89	North America	2	newer
DT18AS72788	Desktop	2018	DT18AS7278801237	Corporate & Enterprise	84	84.8	50.9	0.95	Asia	6	mid-age
LP23NA22027	Laptop	2023	LP23NA2202701237	Corporate & Enterprise	45	22.4	9.0	0.86	North America	1	newer
LP20EU13088	Laptop	2020	LP20EU1308801238	Education & Digital Literacy	53	26.0	6.5	0.87	Europe	4	mid-age
DT22EU83610	Desktop	2022	DT22EU8361001240	Corporate & Enterprise	77	10.5	2.6	0.94	Europe	2	newer
DT20NA18184	Desktop	2020	DT20NA1818401241	Corporate & Enterprise	80	27.2	10.9	0.95	North America	4	mid-age
LP19EU72320	Laptop	2019	LP19EU7232001242	Education & Digital Literacy	52	51.8	12.9	0.87	Europe	5	mid-age
LP22NA45129	Laptop	2022	LP22NA4512901243	Corporate & Enterprise	46	7.5	3.0	0.87	North America	2	newer
LP22AS93395	Laptop	2022	LP22AS9339501244	Corporate & Enterprise	46	22.7	13.6	0.88	Asia	2	newer
DT17AS10160	Desktop	2017	DT17AS1016001247	Corporate & Enterprise	86	62.8	37.7	0.94	Asia	7	older
DT22NA33399	Desktop	2022	DT22NA3339901249	Corporate & Enterprise	78	7.5	3.0	0.96	North America	2	newer
DT20AS67209	Desktop	2020	DT20AS6720901250	Education & Digital Literacy	81	42.8	25.7	0.97	Asia	4	mid-age
LP19NA76792	Laptop	2019	LP19NA7679201253	Corporate & Enterprise	53	37.6	15.0	0.87	North America	5	mid-age
LP21NA36311	Laptop	2021	LP21NA3631101253	Education & Digital Literacy	52	49.9	20.0	0.89	North America	3	newer
LP21NA62353	Laptop	2021	LP21NA6235301253	Education & Digital Literacy	49	30.8	12.3	0.88	North America	3	newer
DT17NA30303	Desktop	2017	DT17NA3030301256	Education & Digital Literacy	86	17.5	7.0	0.95	North America	7	older
LP23AS83877	Laptop	2023	LP23AS8387701259	Education & Digital Literacy	44	15.2	9.1	0.87	Asia	1	newer
LP20NA87065	Laptop	2020	LP20NA8706501261	Corporate & Enterprise	52	5.9	2.4	0.88	North America	4	mid-age
DT17NA53322	Desktop	2017	DT17NA5332201264	Government & Public Sector	84	18.4	7.4	0.94	North America	7	older
LP22NA24736	Laptop	2022	LP22NA2473601264	Government & Public Sector	47	16.7	6.7	0.88	North America	2	newer
DT20AS23425	Desktop	2020	DT20AS2342501264	Government & Public Sector	82	21.0	12.6	0.92	Asia	4	mid-age
LP20NA70286	Laptop	2020	LP20NA7028601265	Corporate & Enterprise	48	17.2	6.9	0.86	North America	4	mid-age
DT16AS74060	Desktop	2016	DT16AS7406901267	Education & Digital Literacy	90	37.7	22.6	0.97	Asia	8	older

-- Aggregate overall impact

```
SELECT
    COUNT(*) AS total_devices,
    AVG(2024 - d.model_year) AS avg_device_age,
    AVG(i.energy_savings_yr) AS avg_energy_savings_kwh,
    SUM(i.co2_saved_kg_yr) / 1000 AS total_co2_saved_tons
FROM intel.device_data d
JOIN intel.impact_data i ON d.device_id = i.device_id;
```

total_devices	avg_device age	avg_energy_savings_kwh	total_co2_saved_tons
601740	3.5229700535114834	25.7420925981320836	6768.4206000000000000

-- Device Type Analysis

```
SELECT d.device_type,
        COUNT(*) AS total_devices,
        AVG(i.energy_savings_yr) AS avg_energy_savings_kwh,
        AVG(i.co2_saved_kg_yr) / 1000 AS avg_co2_saved_tons
FROM intel.device_data d
```

```
JOIN intel.impact_data i ON d.device_id = i.device_id
GROUP BY d.device_type;
```

device_type	total_devices	avg_energy_savings_kwh	avg_co2_saved_tons
Desktop	193676	25.6162931906896053	0.01119045725851421960
Laptop	408064	25.8017997176913425	0.01127543130489335010

```
-- Device Age Analysis
SELECT CASE
        WHEN 2024 - d.model_year <= 3 THEN 'newer'
        WHEN 2024 - d.model_year > 3 AND 2024 - d.model_year <= 6 THEN 'mid-age'
        WHEN 2024 - d.model_year > 6 THEN 'older'
    END AS device_age_bucket,
    COUNT(*) AS total_devices,
    AVG(i.energy_savings_yr) AS avg_energy_savings_kwh,
    AVG(i.co2_saved_kg_yr) / 1000 AS avg_co2_saved_tons
FROM intel.device_data d
JOIN intel.impact_data i ON d.device_id = i.device_id
GROUP BY device_age_bucket;
```

device_age_bucket	total_devices	avg_energy_savings_kwh	avg_co2_saved_tons
mid-age	264310	32.0414842419885740	0.01400321629904279070
newer	317191	19.0713276858422843	0.00832964270739081500
older	20239	48.0215722120658135	0.02100611690300904190

```
-- Regional Analysis
SELECT i.region,
    COUNT(*) AS total_devices,
    AVG(i.energy_savings_yr) AS avg_energy_savings_kwh,
    AVG(i.co2_saved_kg_yr) / 1000 AS avg_co2_saved_tons
FROM intel.device_data d
JOIN intel.impact_data i ON d.device_id = i.device_id
GROUP BY i.region;
```

region	total_devices	avg_energy_savings_kwh	avg_co2_saved_tons
Asia	192881	25.7886774747123874	0.01547317413327388390
Europe	109381	25.7486089905925161	0.00643717556065495840
North America	299478	25.7097092273889902	0.01028400950988052540

```
--Optimizing Repurposing Strategy

WITH regional_totals AS (
   SELECT region,
         SUM(energy_savings_yr) AS total_energy_savings,
         SUM(co2_saved_kg_yr) AS total_co2_saved
   FROM intel.device_data d
   JOIN intel.impact_data i ON d.device_id = i.device_id
   GROUP BY region
)
SELECT i.region,
      d.device_type,
      COUNT(*) AS total_devices,
      AVG(i.energy_savings_yr) AS avg_energy_savings_kwh,
      AVG(i.co2_saved_kg_yr)/1000 AS avg_co2_saved_tons,
      SUM(i.energy_savings_yr) / rt.total_energy_savings * 100 AS pct_energy_savings_by_type,
      SUM(i.co2_saved_kg_yr) / rt.total_co2_saved * 100 AS pct_co2_savings_by_type
FROM intel.device_data d
JOIN intel.impact_data i ON d.device_id = i.device_id
JOIN regional_totals rt ON i.region = rt.region
GROUP BY i.region, d.device_type, rt.total_energy_savings, rt.total_co2_saved;
```

region	device_type	total_devices	avg_energy_savings_kwh	avg_co2_saved_tons	pct_energy_savings_by_type	pct_co2_savings_by_type
Asia	Desktop	61892	25.6795627867898921	0.0154076181089640018	31.95241016151134609900	31.95222901882481220400
Asia	Laptop	130989	25.8402339127713014	0.0155041492033682218	68.04758983848865390100	68.04777098117518779600
Europe	Desktop	35156	25.6104363408806463	0.0064025457958812152	31.96839052401700520300	31.96795874249951747200
Europe	Laptop	74225	25.8140532165712361	0.0064535776355675312	68.03160947598299479700	68.03204125750048252800
North America	Desktop	96628	25.5778987457051786	0.0102312662996232976	32.10005418149453828300	32.09999653877516669200
North America	Laptop	202850	25.7724974118807000	0.0103091338427409416	67.89994581850546171700	67.90000346122483330800